CIVICLENS · TECHNICAL SPECIFICATION

The projection model

Every equation that turns a policy lever into a projected census figure, the parameters behind them, and an explicit ledger of which numbers come from research and which are assumptions of this project.

Document	Mathematical specification of `src/model`
Applies to	Branch `feat/policy-model`, commit `9007f21`
Data	ABS `ABS_REGIONAL_ASGS2021`, reference year 2021
Geography	Statistical Area Level 2, Victoria; state baseline ASGS code 2
Status	First-order projection. Not a forecast.

1 Scope, and what this model is not

CivicLens takes a sustained percentage change in a policy lever — school funding per student, say — and projects it onto the census figures for one Victorian suburb. This document specifies that computation exactly: every function, every parameter, and every assumption interposed between a published research finding and a number on screen.

Three properties bound what the output can mean.

It is first-order. Each effect estimates one lever's own influence on one metric. Effects are summed, never chained. Nothing propagates from education to income to employment, because compounding first-order estimates through a causal chain multiplies small numbers into confident nonsense.

It is a range, never a point. Every projected figure is a triple $(\mathrm{low},\ \mathrm{central},\ \mathrm{high})$. The central value is the source study's own estimate; the bounds are described in §9, which states for each one whether it is a reported figure or a judgement of this project.

It is bounded by evidence, not by ambition. A lever–metric link with a known direction but no transferable effect size is typed so that it is structurally incapable of contributing a number. A link with no research at all has no entry. Of the four levers modelled, one produces quantified projections.

> **THE CENTRAL CLAIM THIS MODEL MAKES**
>
> A census metric counts the whole adult population, but a school funding change reaches one school cohort at a time. The exposure function of §4 is what encodes this, and it is the reason a 20% funding increase moves Year 12 completion by about 2.4 points over twenty years rather than 19. Without it the model would assert that a policy rewrites the schooling of every adult alive the moment it passes.

2 Notation

Symbols used throughout. Percentages are expressed as numbers out of 100, not fractions.

Symbol	Meaning	Domain and notes
m	A census metric	One of eight; see §3
j	A policy lever	One of four; see §8
b_m	Baseline value of metric m	From the ABS; units vary by metric
ℓ_j	Change in lever j	Percent, $\ell_j \in [-50, 50]$
t	Projection horizon	Years ahead, $t \geq 0$
e	An effect: one lever acting on one metric	See §5
L_e	Lag of effect e	Years before any of it can appear
P_e	Phase-in of effect e	Years from the end of the lag to full reach
p_e	Reference lever change	Percent; the change the magnitude is stated for
$g_e^-,\ g_e^0,\ g_e^+$	Magnitude band of e	Low, central, high; units set by κ_e
κ_e	Kind of effect	$\{\mathsf{relative},\ \mathsf{absolute}\}$
E	Exposure function	$\to [0, 1]$; see §4

3 The input layer

Baselines are read from the ABS Data API for one SA2 and, separately, for Victoria as a whole. Seven metrics are reported directly. One is derived, because the API reports it as a head count and a head count cannot be compared between areas of different sizes:

$$b_{\mathsf{bornOverseasShare}} \;=\; 100 \cdot \frac{n_{\mathsf{TOTMIG\_4}}}{n_{\mathsf{ERP\_P\_20}}}$$

defined only when the population measure is present and non-zero. The ABS suppresses measures for small or unusual areas, so the baseline is a *partial* vector: any metric may be absent, and an absent metric is projected as absent rather than as zero.

The eight metrics, their units, and the bounds used when clamping a projection (§6).

Metric	ABS measure	Format	Bounds
Population	ERP_P_20	count	$[0, \infty)$
Median age	ERP_23	years	$[0, \infty)$
Median equivalised household income	EQUIV_2	currency, weekly	$[0, \infty)$
Unemployment rate	LF_4	percent	$[0, 100]$
Median weekly rent	RENT_4	currency	$[0, \infty)$
Residents born overseas	TOTMIG_4 ÷ ERP_P_20	percent	$[0, 100]$
Year 12 completion	HIGH_2	percent	$[0, 100]$
Non-school qualifications	SCHOOL_2	percent	$[0, 100]$

4 The exposure function

This is the heart of the model. Let t be the horizon in years, L the lag and P the phase-in. Exposure is the share of the population counted by a metric that has actually lived through the policy change by time t:

$$E(t;\, L,\, P) \;=\; \begin{cases} 0, & t \le L, \\ 1, & t > L \text{ and } P \le 0, \\ \min\left(\dfrac{t-L}{P},\; 1\right), & t > L \text{ and } P > 0. \end{cases}$$

It is continuous, non-decreasing in t, zero on $[0, L]$, and reaches 1 at $t = L + P$. It has three parts, and each carries a distinct meaning:

- The **lag** L is the delay before the first affected person can appear in the measure. For school funding and Year 12 completion this is 13 years: a child starting school today does not finish Year 12 sooner than that.
- The **phase-in** P is population turnover — how long it takes exposed cohorts to replace the population the census counts. For an adult-population stock this is on the order of half a century.
- The **cap** at 1 states that no more than the whole population can be exposed.

Worked instances of the function, using the parameters of §8:

Effect	L	P	$E(10)$	$E(20)$	$E(30)$	$E(50)$
School funding → Year 12 completion	13	55	0	0.127	0.309	0.673

Effect	L	P	$E(10)$	$E(20)$	$E(30)$	$E(50)$
School funding → household income	18	45	0	0.044	0.267	0.711
School funding → non-school quals	18	50	0	0.040	0.240	0.640

> **STATUS OF THESE PARAMETERS**
>
> L and P are structural assumptions of CivicLens about how long a change takes to reach a census measure. They are *not* findings of the cited studies, none of which estimates a population-turnover profile. The interface labels them as assumptions wherever it shows them.

5 One effect's contribution

An effect e links lever $j(e)$ to metric $m(e)$ and carries a magnitude band (g_e^-, g_e^0, g_e^+) stated for a reference lever change p_e. Define the *scale factor*

$$s_e(t) \;=\; \frac{\ell_{j(e)}}{p_e} \;\cdot\; E(t;\, L_e,\, P_e)\,.$$

The first term extrapolates linearly from the change the study observed to the change the user asked for; the second discounts it by how much of the population has been reached. A magnitude g becomes a change in the metric's own units through

$$\delta_e(g,\, t) \;=\; \begin{cases} b_{m(e)} \cdot \dfrac{g}{100} \cdot s_e(t), & \kappa_e = \textsf{relative}, \\ g \cdot s_e(t), & \kappa_e = \textsf{absolute}. \end{cases}$$

A **relative** magnitude is a percentage of the metric's own value — a 7.25% rise in income. An **absolute** magnitude is a change in percentage points — 9.5 points more Year 12 completion. The two are not interchangeable, and the distinction is carried in the data rather than inferred.

The contribution of e is then the band

$$\Delta_e(t) \;=\; \Big(\underbrace{\min\{\delta_e(g_e^-),\, \delta_e(g_e^+)\}}_{\text{low}},\; \underbrace{\delta_e(g_e^0)}_{\text{central}},\; \underbrace{\max\{\delta_e(g_e^-),\, \delta_e(g_e^+)\}}_{\text{high}}\Big).$$

The minimum and maximum are taken rather than assumed because δ_e is order-reversing when $\ell_{j(e)} < 0$: cutting a lever maps the low magnitude to the high outcome. Since the parameter table satisfies $g^- \le g^0 \le g^+$ — an invariant asserted by the test suite — the central value always lies within the band.

6 Aggregation and bounds

For a metric m, let $A_m(\boldsymbol{\ell})$ be the set of effects targeting m whose lever has moved:

$$A_m(\boldsymbol{\ell}) \;=\; \{\, e \;:\; m(e) = m \;\wedge\; \ell_{j(e)} \neq 0 \,\}, \qquad Q_m = \{\, e \in A_m : e \text{ is quantified} \,\}.$$

Contributions are summed componentwise over the quantified effects only,

$$\Sigma_m(t) \;=\; \sum_{e \in Q_m} \Delta_e(t),$$

and the projection is the baseline plus that sum, clamped componentwise into the metric's admissible range $[\alpha_m, \omega_m]$ from §3:

$$\Pi_m(t) \;=\; \operatorname{clamp}\big(b_m + \Sigma_m(t),\ \alpha_m,\ \omega_m\big), \qquad \operatorname{clamp}(x, \alpha, \omega) = \min\big(\max(x, \alpha),\ \omega\big).$$

Clamping is what stops a percentage exceeding 100 or a rent going negative under a large lever change at a long horizon. It is applied to each of the three band components independently, so a clamped band can be narrower than the unclamped one — the correct behaviour, since the bound is real information about the metric.

Three cases are reported separately rather than collapsed into the number:

- $A_m = \varnothing$: nothing the model knows about touches this metric. The interface says so.
- $Q_m = \varnothing$ but $A_m \neq \varnothing$: every active link is direction-only. $\Sigma_m = 0$ by construction, so the projection equals the baseline, and the direction is reported without a figure.
- $Q_m \neq \varnothing$: a range is projected, badged with the weakest evidence tier among its contributions.

7 The projected series

The chart samples the same function across the horizon. With $K = \max(\operatorname{round}(T),\ 1)$ steps for a horizon T,

$$\tau_k \;=\; \frac{k}{K}\,T, \qquad k = 0, 1, \ldots, K, \qquad \text{giving the points } \big(\tau_k,\ \Pi_m(\tau_k)\big).$$

For integer horizons this is one sample per year. Because Π_m is evaluated by the same function used for the headline figure, the last point of the series is identical to the projection at the horizon; the test suite asserts this rather than trusting it.

The shape of the series is the model's timing made visible: constant at b_m for $\tau \leq \min_e L_e$, then fanning out, with the spread growing as exposure grows.

8 The parameter table

These are all of the numbers. Every entry below is data, not code, and each carries its source in the interface.

Quantified effects. Magnitudes are stated per $p_e = 10\%$ sustained change in the lever.

Lever → metric	Kind	g^-	g^0	g^+	L	P	Evidence
School funding → Year 12 completion	absolute (pp)	3.0	9.5	11.6	13	55	Moderate
School funding → median household income	relative (%)	2.0	7.25	9.5	18	45	Strong
School funding → non-school qualifications	absolute (pp)	1.5	6.0	9.0	18	50	Moderate

Direction-only effects. These carry a sign and a citation and cannot contribute a number.

Lever → metric	Direction	L	P	Why no magnitude
School funding → unemployment rate	down	18	45	The study measures adult poverty, not an area unemployment rate
Employment services → unemployment rate	down	2	5	The meta-analysis reports direction and timing, no transferable effect size
New housing supply → median rent	down	1	3	Estimated on monthly listings in Munich; fades within a year

Health funding has no entry in either table. It is a lever the interface offers because it is a thing governments promise, and moving it changes nothing, because no study links it to any of the eight metrics closely enough to put a number on it.

Beyond $|\ell_j| > 25$ the interface warns that the model is extrapolating linearly past any change a study observed. The linear term ℓ/p has no empirical support at that distance; it is a modelling convenience.

9 Worked example

Alfredton (SA2 201011001), 2021 baselines, with school funding raised 20% and every other lever at zero. This reproduces the figures the interface shows.

9.1 Year 12 completion at a 20-year horizon

Baseline $b = 61.4\%$, parameters $L = 13$, $P = 55$, $p = 10$, band $(3,\ 9.5,\ 11.6)$ in percentage points.

$$E(20;\ 13,\ 55) = \frac{20-13}{55} = \frac{7}{55} = 0.12727, \qquad s = \frac{20}{10} \cdot 0.12727 = 0.25455.$$

The effect is absolute, so $\delta(g) = g \cdot s$:

$$\delta(3) = 0.764, \qquad \delta(9.5) = 2.418, \qquad \delta(11.6) = 2.953 \text{ percentage points.}$$

$$\Pi(20) = 61.4 + (0.764,\ 2.418,\ 2.953) = (62.16,\ 63.82,\ 64.35) \ \longrightarrow\ \mathbf{62.2\%\ to\ 64.4\%}.$$

9.2 The same lever at 50 years

$$E(50) = \frac{37}{55} = 0.67273, \qquad s = 1.34545, \qquad \Pi(50) = (65.44,\ 74.18,\ 77.01) \ \longrightarrow\ \mathbf{65.4\%\ to\ 77.0\%}.$$

Exposure has risen from 13% to 67% of the counted population, and the projection widens accordingly. The whole difference between the two answers is the exposure term.

9.3 Median household income at 20 years

Baseline $b = \$1{,}113$, $L = 18$, $P = 45$, band $(2,\ 7.25,\ 9.5)$ as *relative* percentages.

$$E(20;\ 18,\ 45) = \frac{2}{45} = 0.04444, \qquad s = 0.08889, \qquad \delta(g) = 1113 \cdot \frac{g}{100} \cdot s.$$

$$\delta(2) = 1.98, \quad \delta(7.25) = 7.17, \quad \delta(9.5) = 9.40 \implies \Pi(20) = (1114.98,\ 1120.17,\ 1122.40) \longrightarrow \mathbf{\$1{,}115\ to\ \$1{,}122}.$$

The range is narrow not because the effect is small but because only 4.4% of the counted population has been reached: the lag is 18 years and the horizon is 20.

9.4 Non-school qualifications at 20 years

$$E(20;\ 18,\ 50) = 0.04, \quad s = 0.08, \quad \Pi(20) = 63.2 + (0.12,\ 0.48,\ 0.72) \longrightarrow \mathbf{63.3\%\ to\ 63.9\%}.$$

10 The direct adjustment, which is not part of this model

A second panel scales a figure by itself:

$$\sigma(v,\ c)\ =\ v\left(1 + \frac{c}{100}\right), \qquad c \in [-20,\ 20].$$

It applies to two metrics, asserts no causal relationship, and is deliberately kept apart from the projection: Π_m is always computed from the ABS baseline b_m, never from $\sigma(b_m, c)$. The two panels therefore cannot be mistaken for a single compounding scenario.

11 Comparison against the state

Each metric is also compared against Victoria, computed from the same dataflow at ASGS code 2. For a displayed value v and state baseline β, with $\mathcal{F}$ the display formatter of §13:

$$\mathrm{delta}(v, \beta) = \begin{cases} \text{“Same as Victoria”}, & \mathcal{F}(v) = \mathcal{F}(\beta), \\ \mathcal{F}\big(|v - \beta|\big)\ \text{above / below}, & \text{otherwise.} \end{cases}$$

Equality is tested on the *formatted* strings, not the floats. An exact float comparison would report a difference of “0.0 pp” between two numbers the reader sees as identical; comparing what is displayed makes that impossible.

Tone is a function of direction, not of sign. Each metric declares whether higher is better, lower is better, or neither:

$$\text{tone} = \begin{cases} \mathsf{neutral}, & \text{direction} = \mathsf{neutral}, \\ \mathsf{positive}, & (\text{direction} = \mathsf{higher}) \Leftrightarrow (v > \beta), \\ \mathsf{negative}, & \text{otherwise.} \end{cases}$$

Unemployment declares lower-is-better; Year 12 completion, non-school qualifications and income declare higher-is-better. Median age, rent and the born-overseas share declare neither, because calling a direction “better” there would be an editorial judgement rather than a measurement. Population declares itself incomparable altogether: a suburb count against a state total is not a comparison, so no delta is computed.

12 Chart geometry

The fan chart is drawn from the series of §7 in a 320×132 viewBox with padding $(\mathrm{top}, \mathrm{right}, \mathrm{bottom}, \mathrm{left}) = (10, 12, 24, 48)$, giving a plot of 260×98.

12.1 Vertical domain

Let $V = \{b\} \cup \bigcup_k \{\mathrm{low}_k,\ \mathrm{high}_k\}$, $v_{\min} = \min V$, $v_{\max} = \max V$, and $r = v_{\max} - v_{\min}$. The padding is

$$\rho \ = \ 0.12 \cdot \begin{cases} r, & r > 0, \\ 0.1\,|b|, & r = 0,\ b \neq 0, \\ 1, & \text{otherwise}, \end{cases} \qquad \mathrm{domain} = [\,v_{\min} - \rho,\ v_{\max} + \rho\,].$$

The fallbacks matter: a projection that has not yet left the lag is perfectly flat, and without them the plot would collapse to zero height.

12.2 Coordinate transforms

$$x(\tau) = 48 + \frac{\tau}{T} \cdot 260, \qquad y(v) = 10 + \left(1 - \frac{v - v'_{\min}}{v'_{\max} - v'_{\min}}\right) \cdot 98,$$

where v' denotes the padded domain bounds. The band is a closed path traversing the high edge forward and the low edge in reverse; the central line is a polyline over the same abscissae.

12.3 Axis ticks

Ticks land on round numbers rather than wherever the data ends. For a target count $n = 4$ over a domain of span Δ:

$$\mathrm{rough} = \frac{\Delta}{n-1}, \qquad \mu = 10^{\lfloor \log_{10} \mathrm{rough} \rfloor}, \qquad \mathrm{step} = \min\bigl\{\, c\mu \ :\ c \in \{1, 2, 2.5, 5, 10\},\ c\mu \geq \mathrm{rough} \,\bigr\},$$

and ticks are placed at $\lceil v'_{\min}/\mathrm{step} \rceil \cdot \mathrm{step}$, incrementing by the step while $\leq v'_{\max} + 10^{-9}$. The epsilon absorbs floating-point drift so a tick that lands exactly on the top of the domain is not dropped.

13 Rounding and display

No rounding happens inside the model. Π_m is carried at full precision and rounded exactly once, on the way to the screen:

Format	Rule	As a value	As a difference
count	0 decimals, grouped	16,841	16,841
currency	AUD, 0 decimals	$1,113	$39
percent	1 decimal	61.4%	1.3 pp
years	1 decimal	34.2 years	3.7 years

A difference between two percentages is reported in percentage points, never in percent — the two are different quantities and conflating them is a common way to overstate a change by an order of magnitude.

14 Assumptions ledger

Every number in §8, classified. This table is the honest core of the document: it separates what research measured from what this project decided.

F = finding of the cited study. D = derived from a finding by a stated conversion. A = assumption of CivicLens, with no study behind it.

	Quantity	Value	Basis
F	Graduation effect, all children	+9.5 pp per 10%	Jackson, Johnson & Persico (2016), spending sustained across all 12 school years
F	Graduation effect, low-income children	+11.6 pp per 10%	Same study; used as the high bound
A	Year 12 low bound	+3.0 pp	A conservative floor for carrying a United States estimate to Victoria. Not a reported confidence bound.
F	Adult wage effect	+7.25% per 10%	Same study, all children
F	Adult wage effect, low-income	+9.5% per 10%	Same study; used as the high bound
A	Income low bound	+2.0%	Conservative transfer floor, as above
F	College-going effect	+2.8 pp per $1,000 per pupil sustained four years	Jackson & Mackevicius (2024) meta-analysis
D	Non-school qualifications central	+6.0 pp per 10%	Scaled by $21,550 per FTE student (RoGS 2026): $2.8 \times 2.155 \approx 6.0$. Assumes linear scaling in dollars and that college-going stands in for holding a qualification.
A	Non-school qualifications bounds	1.5 and 9.0 pp	Judgement, widened to reflect the two conversions above
A	All lags L	13, 18, 18, 2, 1	Structural: time for a first affected person to reach the measure
A	All phase-ins P	55, 45, 50, 5, 3	Structural: population turnover in the measure
A	Linearity in the lever	ℓ/p	Studies observed roughly 10% changes; ±50% is available and flagged past 25%
A	Additivity across levers	$\sum_e \Delta_e$	Chosen over chaining; no interaction terms are estimated
A	Cohort exposure is linear in time	$(t-L)/P$	A ramp, not a demographic turnover model
A	Transferability of US estimates	—	The strongest evidence here is American; the low bounds respect this but cannot remove it

15 Limitations

- **The census contains no policy levers.** There is no funding column at SA2. Levers are exogenous inputs, and the link to any census outcome comes entirely from external research.
- **Cross-sectional correlation is not used, and could not be.** Fitting the relationship between attainment and income across SA2s would be an ecological inference over confounded units. The model deliberately imports causal estimates instead.
- **Unit conversions are unverified.** Two of the three quantified links convert what a study measured — a cohort graduation probability, college-going — into an adult-population stock. Neither study tests that conversion.

- **No migration response.** Nothing models people moving into or out of a suburb because of a policy, which is one of the larger real-world channels for an area-level effect.
- **No interaction, no diminishing returns.** Two levers moved together add; a lever moved twice as far does twice as much.
- **Exposure is a ramp.** Real cohort replacement is not linear, and the phase-ins are round numbers rather than estimates.
- **One reference year.** All baselines are 2021, so the starting point ages while the horizon does not.

16 Where each equation lives in the source

Section	Function	File
§3 derived share	`deriveDemographics`	`src/utils/fetchDemographics.ts`
§4 exposure	`exposure`	`src/model/project.ts`
§5 contribution	`contributionBand`	`src/model/project.ts`
§6 aggregation	`projectOutcome`	`src/model/project.ts`
§7 series	`projectSeries`	`src/model/project.ts`
§8 parameters	`EFFECTS`	`src/model/effects.ts`
§10 direct adjustment	`simulate`	`src/utils/simulate.ts`
§11 comparison	`createDelta`	`src/utils/format.ts`
§12 chart geometry	`buildScales`, `niceTicks`	`src/components/ProjectionChart.tsx`
§13 display	`formatValue`	`src/utils/format.ts`

17 References

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